

Mathematical Methods for Scientists and Engineers

Problem Set 1

Due: October 6, 2026

Throughout, bold letters denote vectors, $A = |\mathbf{A}|$ is the magnitude of $\mathbf{A}$, $\mathbf{r} = x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}}$ is the position vector, $r = |\mathbf{r}|$, and $\hat{\mathbf{r}} = \mathbf{r}/r$.

Problem 1:

Three vectors form a triangle, $\mathbf{A} = \mathbf{B} - \mathbf{C}$, and let θ be the angle between $\mathbf{B}$ and $\mathbf{C}$. Starting from $A^2 = (\mathbf{B} - \mathbf{C})^2$, prove the law of cosines, $A^2 = B^2 + C^2 - 2BC \cos \theta$.

Problem 2:

Prove that $(\mathbf{A} \times \mathbf{B}) \cdot (\mathbf{A} \times \mathbf{B}) = (AB)^2 - (\mathbf{A} \cdot \mathbf{B})^2$.

Hint. Use the geometric definitions $|\mathbf{A} \times \mathbf{B}| = AB \sin \theta$ and $\mathbf{A} \cdot \mathbf{B} = AB \cos \theta$. What does the result say about the area of the parallelogram spanned by $\mathbf{A}$ and $\mathbf{B}$?

Problem 3:

Verify the expansion of the triple vector product (the “BAC–CAB” rule)

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$$

by direct expansion in Cartesian coordinates.

Hint. It is enough to check the x -component carefully; the y - and z -components follow by cyclic relabeling $x \rightarrow y \rightarrow z \rightarrow x$.

Problem 4:

A vector $\mathbf{D}$ is written as a linear combination of three noncoplanar (but not necessarily orthogonal) vectors:

$$\mathbf{D} = a\mathbf{A} + b\mathbf{B} + c\mathbf{C}.$$

Show that the coefficients are given by ratios of triple scalar products,

$$a = \frac{\mathbf{D} \cdot (\mathbf{B} \times \mathbf{C})}{\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})}, \quad b = \frac{\mathbf{A} \cdot (\mathbf{D} \times \mathbf{C})}{\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})}, \quad c = \frac{\mathbf{A} \cdot (\mathbf{B} \times \mathbf{D})}{\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})}.$$

Hint. Take the dot product of both sides with $\mathbf{B} \times \mathbf{C}$. Why is the denominator nonzero?

Problem 5:

Show that $\nabla(uv) = v\nabla u + u\nabla v$, where u and v are differentiable scalar functions of x , y , and z .

Problem 6:

Prove $\nabla \cdot (\mathbf{a} \times \mathbf{b}) = \mathbf{b} \cdot (\nabla \times \mathbf{a}) - \mathbf{a} \cdot (\nabla \times \mathbf{b})$.

Hint. Treat the left-hand side as a triple scalar product and remember that ∇ is a differential operator: by the product rule it acts once on $\mathbf{a}$ and once on $\mathbf{b}$. You may also verify it by writing out the components.

Problem 7:

The vector potential $\mathbf{A}$ of a magnetic dipole with constant dipole moment $\mathbf{m}$ is

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \frac{\mathbf{m} \times \mathbf{r}}{r^3}.$$

Show that, for $r \neq 0$, the magnetic field $\mathbf{B} = \nabla \times \mathbf{A}$ is

$$\mathbf{B} = \frac{\mu_0}{4\pi} \frac{3\hat{\mathbf{r}}(\hat{\mathbf{r}} \cdot \mathbf{m}) - \mathbf{m}}{r^3}.$$

Hint. Useful intermediate results: $\nabla r = \hat{\mathbf{r}}$, $\nabla \cdot \mathbf{r} = 3$, and $\nabla(r^{-3}) = -3\mathbf{r}/r^5$. This is the familiar dipole field of a bar magnet or a small current loop.

Problem 8:

Consider the force $\mathbf{F} = r^{2n} \mathbf{r}$, where n is a real constant. Find a) $\nabla \cdot \mathbf{F}$, b) $\nabla \times \mathbf{F}$, c) a scalar potential $\phi(x, y, z)$ such that $\mathbf{F} = -\nabla\phi$, d) the value of the exponent n for which the scalar potential diverges at both the origin and infinity.

Hint. For part c), the force is radial and depends only on r , so try $\phi = \phi(r)$ and use $\nabla\phi(r) = \phi'(r)\hat{\mathbf{r}}$. Treat one value of n separately.